

Concavity of van der Waerden numbers

OPG-404

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

For k and ℓ positive integers, the (mixed) van der Waerden number $w(k, \ell)$ is the least positive integer n such that every (red-blue)-coloring of $[1, n]$ admits either a k -term red arithmetic progression or an ℓ -term blue arithmetic progression.

Conjecture For all k and ℓ with $k \geq \ell$, $w(k, \ell) \geq w(k + 1, \ell - 1)$.